EXECUTIVE DECISION REPORT

Retail Business Intelligence | Revenue, Customer & Commercial Performance

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Reporting window: 01 Jan 2025 – 09 Nov 2026
Project: Data Analytics & BI Capstone

DATA STATUS: SYNTHETIC DEMONSTRATION
This report uses generated sample transactions because the supplied source is a data dictionary, not actual sales records. These figures are fictional and must not be represented as company results.

Decision objective: demonstrate a reproducible BI workflow that turns order records into management insights, prioritised actions and measurable pilot hypotheses.

1. Executive Summary

Recognized revenue: ■5,787,322 **Paid orders:** 940 **Average revenue per paid order:** ■6,157 **Paid-order share:** 78.3%

The largest category in this generated sample is **Course Access** (63.4% of recognized revenue). The largest segment is **Student** (40.0%).

• Monitor payment states separately from realized revenue.

• Prioritise category-led merchandising experiments.

• Introduce discount governance and measure contribution margin before scaling promotions.

All metrics above are synthetic demonstration outputs, not actual sales results.

2. Business Context & Decision Questions

The model represents a retail/learning-products business with Learning Kit, Course Access and Mentor Session categories. The data dictionary specifies order ID/date, segment, city, category, quantity, unit price, discount and payment status.

- Which categories contribute most to paid revenue?
- Which customer segments and cities represent the largest revenue pools?
- How do discount levels relate to revenue per paid order?
- How many orders remain Pending, Failed or Refunded?
- Which low-risk experiments should management test first?

Scope boundary: source dictionary only; transaction data in this package is synthetic.

3. Data & Methodology

Method: generate reproducible sample records, validate IDs/dates/quantities/prices/discounts, calculate gross and net revenue, then aggregate by month, category, segment, city and payment status.

• Gross revenue = quantity × unit price.

• Net revenue = gross revenue × (1 – discount percentage / 100).

• Recognized revenue = net revenue for Paid orders only.

• Pending, Failed and Refunded orders are excluded from recognized revenue in this demonstration.

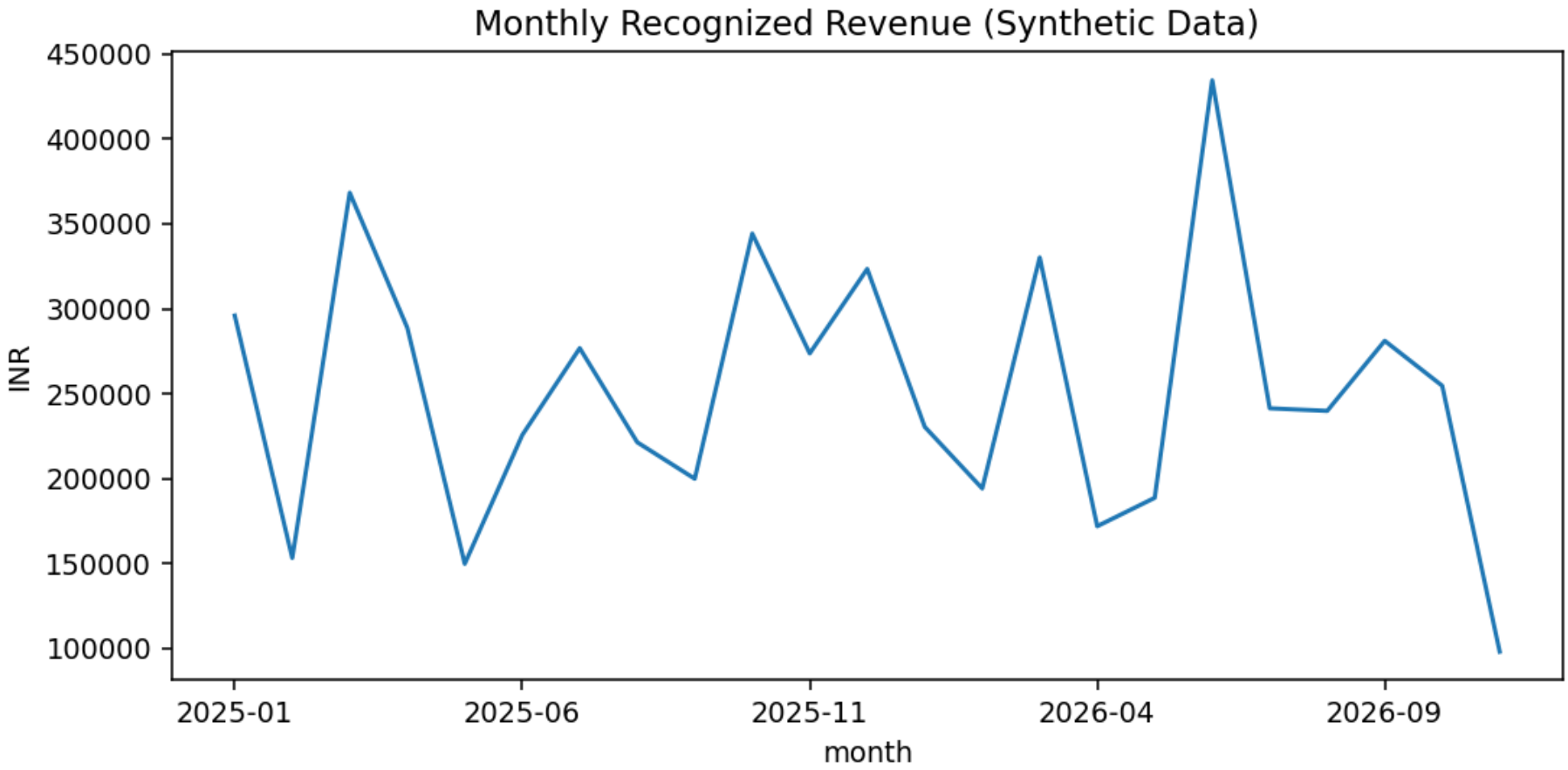
Limitation: the dictionary has no cost or margin fields; profit, contribution margin and true ROI cannot be calculated.

4. KPI Scorecard

Metric	Synthetic sample result
Total orders	1,200
Paid orders	940
Paid-order share	78.3%
Recognized revenue	■5,787,322
Revenue per paid order	■6,157
Average discount on paid orders	9.8%

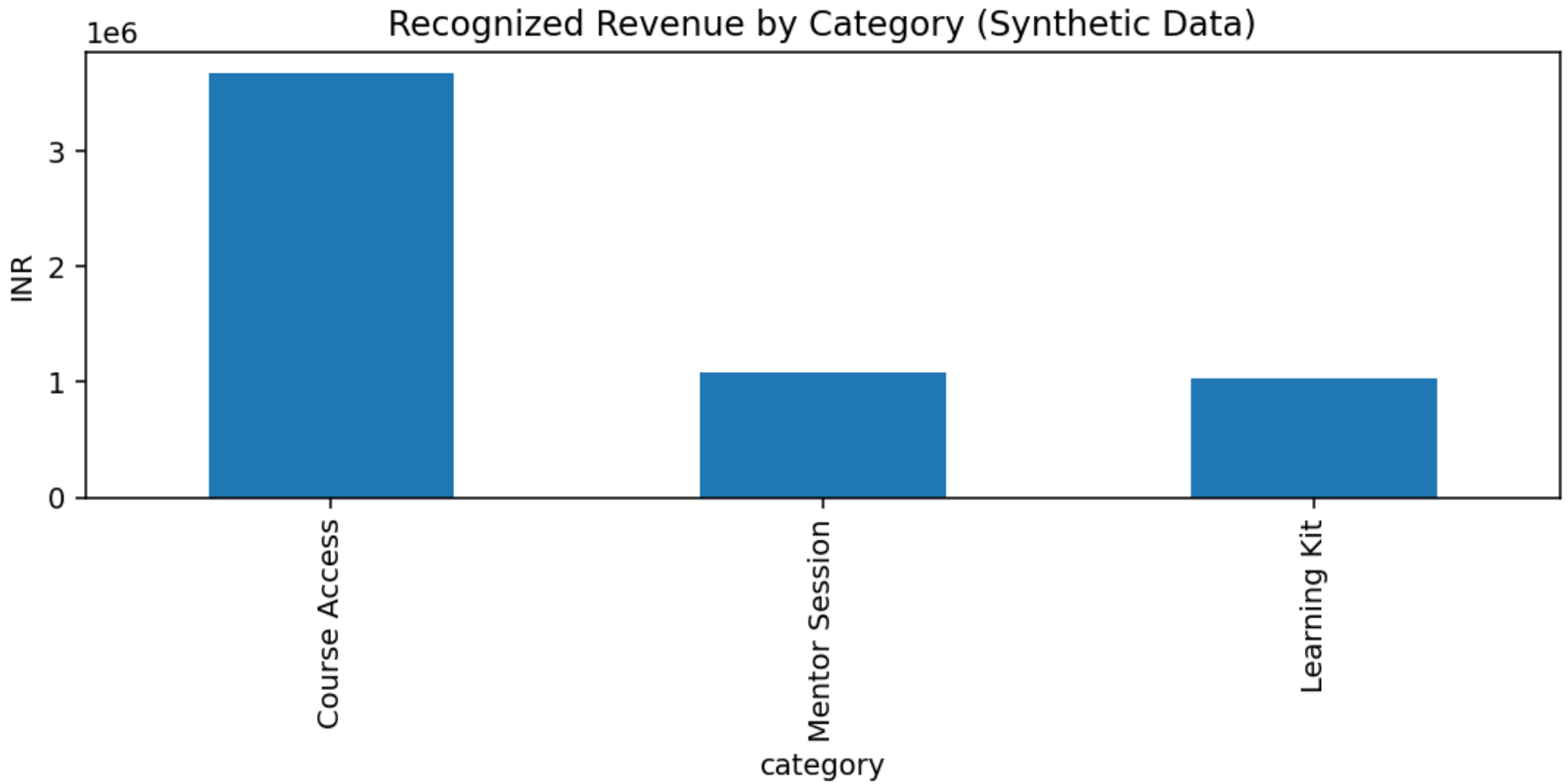
Definitions: recognized revenue sums net revenue for Paid records; paid-order share is Paid orders divided by all orders; average paid-order revenue is recognized revenue divided by Paid order count.

5. Revenue Trend



The monthly trend is intended to reveal seasonality and sustained changes. Generated variation in this sample has no verified business cause. Validate campaigns, calendar effects, stock availability and data completeness before interpreting spikes.

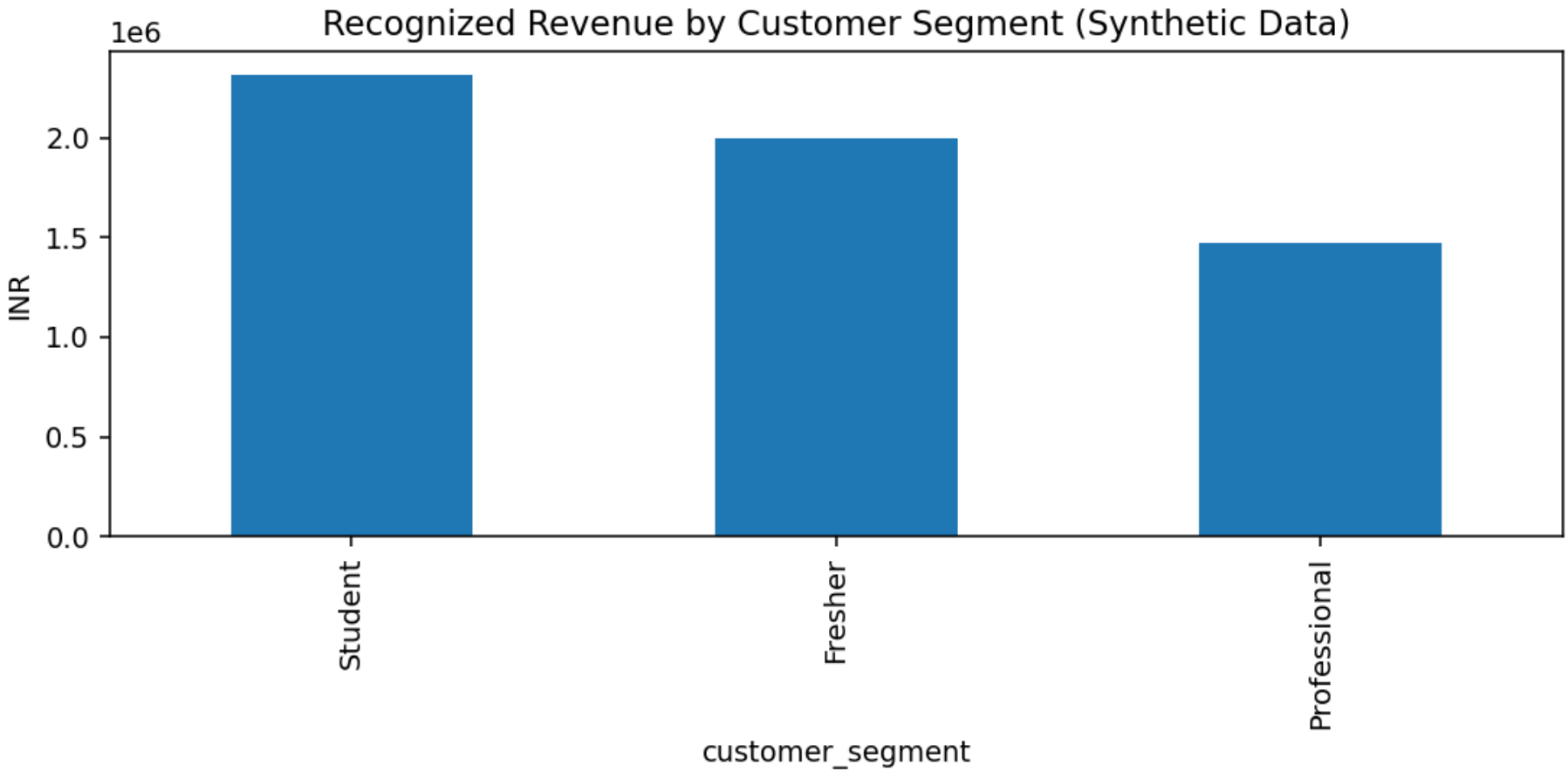
6. Product Category Performance



Category	Recognized revenue (■)	Share
Course Access	3,668,956	63.4%
Mentor Session	1,082,728	18.7%
Learning Kit	1,035,639	17.9%

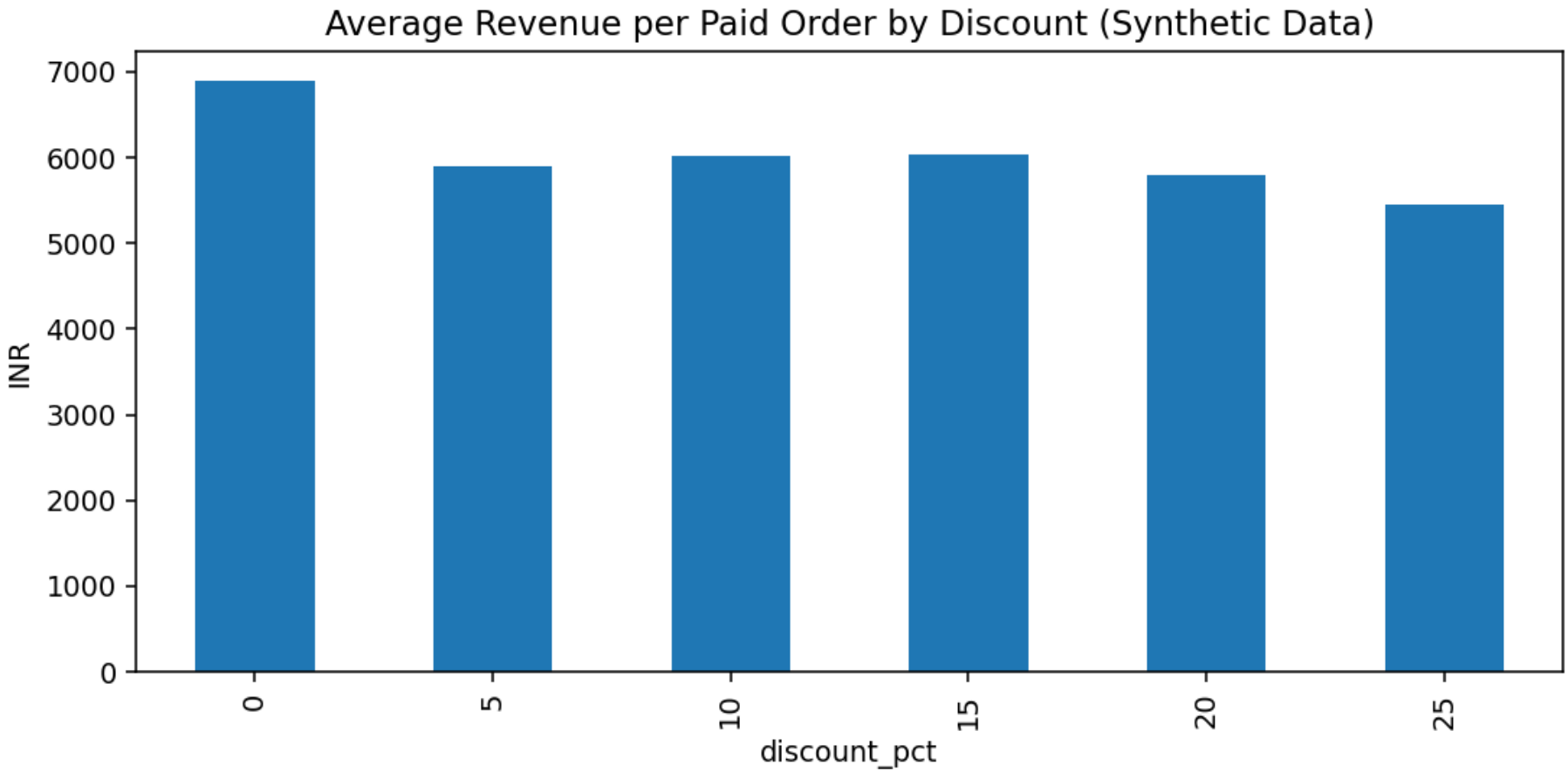
Use category mix to prioritise product-page visibility and bundle tests. Confirm real sales and margin before changing inventory or marketing allocation.

7. Customer & Geographic Mix



Segment revenue indicates the size of paid-revenue pools, not profitability or lifetime value. The notebook also calculates city-level revenue; normalise city comparisons for customer count, order count and marketing spend.

8. Discount & Payment Risk



Discount governance: descriptive comparisons do not prove that discounts cause higher or lower revenue. Use matched cohorts or controlled tests.

Payment status	Orders	Share
Paid	940	78.3%
Pending	109	9.1%

Failed	84	7.0%
Refunded	67	5.6%

Pending/Failed orders need payment-gateway reconciliation before follow-up. Refunded value should be reconciled against settlement records.

9. Key Findings & Risk Register

- Course Access leads sample category revenue (63.4%).
- Student leads sample segment revenue (40.0%).
- 260 orders are not Paid and excluded from recognized revenue.
- Average discount on Paid orders is 9.8%; margin impact is unknown.

Risk	Potential impact	Control
Synthetic data	Not representative	Replace with real order export
No cost/margin fields	Profit and ROI unknown	Add COGS, fees and spend
Payment-state ambiguity	Cash/revenue misstatement	Reconcile gateway settlements
Discount selection bias	Misleading conclusions	Matched cohorts / A-B test
No customer history	No retention/LTV analysis	Add customer ID and cohorts

10. Three Strategic Recommendations

1. Payment recovery workflow

Create a daily Pending/Failed review queue, verify gateway state and track recovery rate. Never trigger duplicate payment collection.

2. Category-led merchandising experiment

A/B test landing-page placement and bundles for the leading category. Measure conversion, net revenue per visitor and refund rate.

3. Discount governance

Set discount approval bands and compare matched cohorts. Add product cost and contribution margin before scaling promotions.

Start with low-cost controls and controlled pilots. Scale only when real data shows positive incremental economics.

11. ROI Scenarios & 90-Day Roadmap

Illustrative planning scenario, not a forecast: if a future validated eligible revenue base were ■10,00,000 annually, a 2%–5% uplift would equal ■20,000–■50,000 incremental annual revenue. This is revenue, not profit.

ROI formula: (Incremental contribution profit – implementation cost) ÷ implementation cost × 100%. True ROI requires margin and cost data.

Period	Workstream	Deliverable
Days 1–15	Data readiness	Validate actual export, definitions, status and cost fields
Days 16–30	Baseline dashboard	Approve KPI definitions and weekly reporting
Days 31–60	Pilot experiments	Payment follow-up, category A/B test, discount guardrails
Days 61–90	Evaluate & scale	Compare test/control groups and incremental contribution

Decision requested: approve data-readiness work and limited pilots; defer broad pricing/budget changes until actual transaction and cost data validate the opportunity.

12. Appendix & Submission Notes

Package contents: Executive report PDF, reproducible Jupyter notebook, HTML dashboard, synthetic transaction CSV, supplied data dictionary and chart images.

- Open the notebook in Jupyter or Google Colab.
- Keep the CSV in the same working folder and run all cells.
- Open retail_dashboard.html in a browser; internet is required for the Chart.js library.
- Replace synthetic records with authorised real data before presenting actual business findings.
- Add cost of goods, fees, marketing spend and implementation cost for genuine ROI analysis.

Prepared as an academic capstone demonstration. All sample values are fictional.